Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam A · Form 1: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. Equal masses share an ideal fixed pulley. Give one side a small downward speed, then release. What happens while the rope stays taut and before either mass hits the floor?

(A) Both stop immediately.

(B) The lighter side rises.

(C) Motion continues at constant speed.

(D) Speed steadily grows.

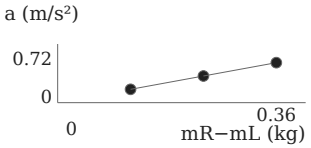
2. Transfer mass between two sides of an ideal Atwood pulley, keeping total mass 5 kg. Right downward is positive; g=10 m/s². mR-mL (kg) -> a (m/s²): (0.12, 0.24); (0.24, 0.48); (0.36, 0.72). What is the signed change in the vertical reading from first to last trial?

(A) 0.672

(B) 0.288

(C) 0.48

(D) 0.864



3. A stationary pulley has two vertical strands. One strand pulls down on the pulley with tension T. What force is Newton's third-law partner of this pull?

(A) The second strand pulling down on the pulley

(B) Earth pulling down on the pulley

(C) The axle pushing upward with force 2T

(D) The pulley pulling up on that same strand with force T

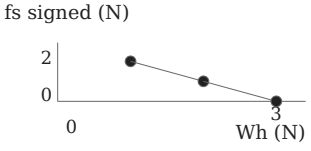
4. A stationary incline block with 3 N of downhill weight is connected to a hanger over an ideal pulley. Static friction is sufficient; count uphill friction positive. Wh (N) -> fs signed (N): (1, 2); (2, 1); (3, 0). As Wh (N) increases, what happens to fs signed (N)?

(A) reversal midway

(B) no change

(C) increase

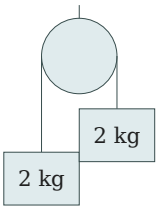
(D) decrease



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Two 2 kg masses hang at rest on an ideal pulley. Use g=10 m/s².

a. Isolate one mass and draw tension upward and weight downward. **(1 pt)**



b. Use equilibrium to determine the tension and the total rope load on the axle. **(3 pt)**

6. A negligible-weight fixed pulley has two vertical rope segments below it. A support holds its axle. T (N) -> Fsupport (N): (1, 2); (2, 4); (3, 6).

a. Build the model for pulley support load from physics and predict both slope and intercept. **(2 pt)**

b. Plot the first and last measurements and describe the trend. **(2 pt)**

7. Two equal masses m hang from a taut ideal rope over a fixed pulley. Show that the acceleration is zero and determine the rope tension. **(3 pt)**